

Hanxi (Gary) Chen

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EDUCATION

Cornell University

Ph.D. in Computer Science, Programming Languages
Advisor: Prof. Alexandra Silva, Prof. Andrew C. Myers

Ithaca, NY
Present

University of Pennsylvania, Penn Engineering & The Wharton School

M.S.E. in Computer and Information Science
B.S.E. in Computer Science, Mathematics minor
B.S. in Economics

Philadelphia, PA
May 2025

Cumulative GPA: 3.97/4.00, *Summa Cum Laude*

Advisor: Prof. Steve Zdancewic, Prof. Sanjeev Khanna

PUBLICATIONS

Hanxi Chen, Noam Zilberstein, Andrew C. Myers, Alexandra Silva. [Oblivious Probabilistic Outcome Logic: Verifying Probabilistic Programs with an Oblivious Adversary](#). arXiv, 2026.

Calvin Beck, Hanxi Chen, Steve Zdancewic. [Vellvm: Formalizing the Informal](#). The Eleventh International Workshop on Coq for Programming Languages (CoqPL), 2025.

Calvin Beck, Hanxi Chen, Steve Zdancewic. [Vellvm: Formalizing the Informal LLVM \(An Experience Report\)](#). NASA Formal Methods: 17th International Symposium (NFM), 2025.

Calvin Beck, Irene Yoon, Hanxi Chen, Yannick Zakowski, Steve Zdancewic. [A Two-Phase Infinite/Finite Low-Level Memory Model: Reconciling Integer-Pointer Casts, Finite Space, and undef at the LLVM IR Level of Abstraction](#). Proceedings of the ACM on Programming Languages, 8 (ICFP), 2024.

Manuscripts

Silei Ren, Suraaj Kanniwadi, Hanxi Chen, Andrew C. Myers. Type-Confusion Security. 2026.

Calvin Beck, Hanxi Chen, Steve Zdancewic. GenLLVM: Randomized Differential Testing for LLVM IR. 2025.

Hanxi Chen. Knapsack Problems with Precedence Constraints. Wharton Undergraduate Thesis, 2024. Supervised by Prof. Sanjeev Khanna.

Hanxi Chen. Random Differential Testing at the LLVM IR Level. CIS Undergraduate Thesis, 2024. Reviewed by Prof. Steve Zdancewic (advisor) and Prof. Benjamin C. Pierce.

RELEVANT RESEARCH EXPERIENCE

Research Assistant, *Cornell PL Group*, advised by Prof. Alexandra Silva and Prof. Andrew C. Myers
Cornell University, Computer Science

Sep 2025 – Present

- Developed Oblivious Probabilistic Outcome Logic (opOL), a logic for probabilistic nondeterministic programs under oblivious adversaries, using schedule consumption and probabilistic independence to verify randomized caching and leader-election protocols.
- Mechanized opOL in Lean 4, including monadic semantics, probability-space constructions, probabilistic assertions, semantic and syntactic outcome triples, soundness proofs, and verified case studies.
- Contributed Lean mechanization for Type-Confusion Security, formalizing static and dynamic information-flow checks for decentralized systems with untrusted components.

Research Mentor, *Cornell Math+AI*, advised by Prof. Daniel Halpern-Leistner
Cornell University, Mathematics

Jan 2026 – Present

- Mentoring autoformalization on research-level mathematical problems and proofs gathered for translation into Lean 4.
- Co-developing tools for evaluating Lean proof quality through proof-structure comparison and converting LaTeX proofs into Lean.

Research Assistant, *PLClub*, advised by Prof. Steve Zdancewic
University of Pennsylvania, Computer and Information Science

Jun 2021 – Aug 2025

- Contributed to Vellvm in Prof. Steve Zdancewic's group, formalizing LLVM IR operational semantics in Rocq, proving semantic lemmas including serialization, and developing low-level memory-model support for pointer-integer casts.
- Built Rocq/OCaml validation tools for testing Vellvm's semantic faithfulness against LLVM implementations, including QuickChick generators for UB-free LLVM programs, interpreter-based traces, and differential tests that found novel LLVM bugs.
- Developed Rocq infrastructure for permutation reasoning and list AC unification, then applied it to formalize first-order classical linear logic with cut elimination and a linear-resource calculus for asynchronous queries.
- Contributed to DeepSpec Software Foundations material presented at the Oregon Programming Languages Summer School 2024.

Research Assistant, Wharton Research Scholars, advised by Prof. Sanjeev Khanna
University of Pennsylvania, The Wharton School Jan 2023 – Dec 2024

- Studied precedence-constrained knapsack problems, which are strongly NP-hard in general, and proved pseudo-polynomial algorithms for structured poset classes including unions of three paths and one-directional bipartite posets with consecutive children.
- Presented the work in the Wharton Research Scholars seminar and wrote an undergraduate thesis.

Research Assistant, Penn Computational Intelligence Lab, advised by Prof. Jing Li
University of Pennsylvania, Electrical and Systems Engineering Feb 2021 – Sep 2021

- Built a C++ simulator prototype for modeling Intel Optane Memory behavior.

HONORS & AWARDS

CIS Faculty Appreciation Award	2024	Eta Kappa Nu	2022, 2023, 2024, 2025
MCIT TA Award	2024	Tau Beta Pi	2023, 2024, 2025
CIS Senior Thesis	2024	Beta Gamma Sigma	2023, 2024, 2025
Wharton Research Scholars	2024	Dean's List	2022, 2023

TEACHING EXPERIENCE

Graduate Teaching Assistant, Formal Verification (CS 45160) 26Sp
 Cornell University. Instructor: Prof. Michael Clarkson.

Graduate Teaching Assistant, Programming Languages and Logics (CS 4110) 25Fa
 Cornell University. Instructor: Prof. Nate Foster.

Teaching Assistant, Compilers and Interpreters (CIS 5521) 25Sp
 University of Pennsylvania. Instructor: Prof. Steve Zdancewic.

Teaching Assistant, Advanced Programming (CIS 5520) 24Fa
 University of Pennsylvania. Instructor: Prof. Stephanie Weirich.
 Taught four lectures and mentored four final-project groups.

Head Teaching Assistant, Analysis of Algorithms (CIS 5020 / CIT 5960 / CIS 3200) 22Su, 22Fa, 23Sp, 23Su, 23Fa, 24Su
 University of Pennsylvania. Instructor: Prof. Sanjeev Khanna.
 Led TA staff and student support across six terms.

Head Teaching Assistant, Theory of Networks (NETS 3120) 24Sp
 University of Pennsylvania. Instructor: Prof. Victor Preciado.

Teaching Assistant, Software Foundations (CIS 5000) 23Fa
 University of Pennsylvania. Instructor: Prof. Benjamin Pierce.
 Taught one lecture on maps in Rocq mechanization.

STEM Tutor, The Weingarten Center 21Sp, 21Su, 21Fa, 22Sp, 22Su
 University of Pennsylvania.
 Tutored core CS and mathematics courses.

COMMUNITY INVOLVEMENT

Artifact Evaluation Committee, ATVA 2026	Jul 2026	Student Volunteer, POPL 2025	Jan 2025
Artifact Evaluation Committee, PLDI 2026	Jan 2026	Student Volunteer, SPLASH 2024	Oct 2024
Artifact Evaluation Committee, POPL 2026	Nov 2025	Student Volunteer, OPLSS 2024	Jun 2024

SELECTED TECHNICAL PROJECTS

RISC-V Pipelined Processor – SystemVerilog, Python, FPGA Feb 2025 – May 2025
University of Pennsylvania, CIS 5710 Computer Organization and Design

- Implemented a RISC-V pipelined processor with AXI-Lite-style caching, multi-cycle execution units, branch prediction, and memory-mapped I/O.
- Built custom verification testbenches, optimized LUT usage and timing behavior, and synthesized the design to FPGA to run compiled programs.

Oat Compiler – *OCaml, C, LLVM IR, x86, Clang, Dune*

Feb 2024 – May 2024

University of Pennsylvania, *CIS 7000 Compilers and Interpreters*

- Implemented a compiler for Oat, a C-like language with pointers and structured data, including lexing, parsing, type checking, frontend lowering, and backend code generation to LLVM IR and x86.
- Added static analyses and optimizations including mem2reg and dead-code elimination, validated against thousands of test cases.

SQL Interpreter – *Haskell, MySQL, Cabal*

Sep 2023 – Dec 2023

University of Pennsylvania, *CIS 5520 Advanced Programming*

- Led development of a Haskell library for parsing, interpreting, optimizing, and pretty-printing SQL, using GADTs, monad transformers, applicative parsers, and property-based testing.
- Designed typed abstract syntax and execution interfaces for local query evaluation.

OTHER EXPERIENCE

Administrator & Lecturer, *Gateways Summer Program*, supervised by Steve Downes

Feb 2025 – Jul 2025

Tabor Academy

- Co-organized a summer program hosting 65 students from Shenzhen Nanshan Foreign Language School across two eight-day sessions, providing exposure to U.S. boarding high school life and STEM courses, including marine science, computer science, and global economies.
- Taught seminars on business negotiation using interactive group activities, covering concepts such as fixed-pie perception, deal breakers, bargaining zones, and practical negotiation strategies.

Research Assistant, *The Knowledge Puzzle Project*, supervised by Prof. Andrew Carton

Feb 2021 – Sep 2021

University of Pennsylvania, The Wharton School

- Conducted research on automating knowledge database construction through PDF data extraction.

Research Intern (Volunteer), *Cybersecurity & Legislation*

Jun 2019 – Aug 2019

Massachusetts State House, House Committee on Technology and Intergovernmental Affairs

- Researched and submitted a report for the Committee hearing regarding telehealth security, analyzing telehealth cybersecurity strategies at the federal and state levels, and suggesting potential improvements for legislation.
- Researched and wrote proposals regarding solutions to the cybersecurity workforce shortage in Massachusetts.